

Digital Clock & Stopwatch

AT89C51 Microcontroller | Embedded C | Proteus Simulation

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1. Project Overview

This project presents a fully functional **Real-Time Digital Clock and Stopwatch** implemented on the **AT89C51** 8-bit microcontroller. The system displays live time on a **LM016L 16×2 LCD** and supports a dedicated stopwatch mode accessible through push buttons. The entire circuit is designed and verified using **Proteus 8** simulation and the firmware is developed in **Keil µVision 5** using Embedded C.

Parameter	Detail
Microcontroller	AT89C51 (8051 Architecture)
Display	LM016L — 16×2 Character LCD
Development IDE	Keil µVision 5
Simulation Tool	Proteus 8 Professional
Programming Lang	Embedded C
Oscillator	11.0592 MHz Crystal

2. Objectives

- Design a real-time clock that accurately tracks and displays HH:MM:SS
- Implement a stopwatch with Start, Stop, and Reset controls
- Use Timer 0 interrupt (ISR) for precise, non-blocking timekeeping
- Interface a 16×2 LCD for clean formatted time display

- Validate the complete circuit through Proteus 8 simulation
- Provide audio feedback via a buzzer on user button interactions

3. Key Features

Feature	Description
Real-Time Clock	Continuously displays hours, minutes, and seconds
Stopwatch Mode	Independent stopwatch with centisecond precision
Mode Switching	Toggle between Clock and Stopwatch via BTN 1 (P1.5)
Start / Stop	Stopwatch controlled by BTN 2 (P1.6)
Reset	Stopwatch resets to 00:00:00 via BTN 3 (P1.7)
Buzzer Feedback	Audio confirmation on every button press
ISR Timekeeping	Timer 0 interrupt ensures non-blocking accuracy
LCD Display	Formatted time output on 16×2 character LCD

4. System Architecture

The system is built around the **AT89C51** microcontroller running at 11.0592 MHz. The firmware is organized into three logical layers:

- **Hardware Abstraction Layer (HAL)** — LCD driver routines for command and data writing
- **Timekeeping Engine** — Timer 0 ISR fires every 1ms, accumulating counts for seconds, minutes, and hours
- **Application Logic** — Main loop polls button states, updates display mode, and formats output string

On startup, the system initializes the LCD, configures Timer 0 in 16-bit mode, enables global interrupts, and enters the main polling loop. The ISR runs independently in the background, ensuring time accuracy is never affected by display or button operations.

5. Tools & Technologies

Tool / Technology	Version / Detail	Purpose
Keil µVision	Version 5	Firmware development & HEX compilation
Proteus	Version 8 Pro	Circuit design & simulation
Embedded C	C89/C90 Standard	Firmware programming language
AT89C51	8051 Architecture	Main microcontroller
LM016L LCD	16×2 Character	Time display output